



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 31.07.2026

In addition to part I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No. : PHY F311

Course Title : QUANTUM MECHANICS II

Instructor-in-Charge : ARPAN DAS

1. Course Description:

This course is the second of the two-semester course on Quantum Mechanics and is a continuation of Quantum Mechanics I. In this course we will discuss quantum theory of angular momentum and addition of angular momentum, applications to spherically symmetric potential, approximation methods to time independent systems, and time dependent systems.

2. Scope and Objective of the Course:

- Quantum theory of angular momentum.
- Three-Dimensional Problems in Quantum Mechanics, Central Potential problem
- Time independent approximation methods in quantum mechanics
- Quantum mechanics of time-dependent systems and quantum theory of scattering.

3. Textbooks:

TB: Modern Quantum Mechanics, J.J. Sakurai & Jim Napolitano, Second Edition, Addison-Wesley Publishing

4. Reference Books:

RB1: Principles of Quantum Mechanics, R Shankar (Second Edition), Springer.

RB2: Quantum Mechanics B.H. Bransden, C. J. Joachain, (second edition) Pearson.

5. Course Plan:

Lecture No.	Learning Objectives	Topics to be Covered	Learning Outcome	Reference to Text and Reference Book
1 – 6	Angular Momentum	General Formalism of Angular Momentum, Matrix Representation of Angular Momentum, Eigenfunctions of Orbital Angular Momentum. Addition of Angular Momenta.	Learn quantum theory of angular momentum, angular momentum algebra.	TB: Ch. 3.



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7 – 17	Three-Dimensional Problems, Central Potential problem	Schrödinger Equation in Spherical Coordinates, General structure of eigen functions in central potential, Hydrogen atom problem.	To be able to write down the radial part of Schrodinger equation for central potentials and apply them to some exactly solvable problems.	TB: Ch. 3, RB1: Ch. 13
18-28	Time independent approximation methods	Time independent perturbation – non-degenerate and degenerate cases, simple applications, the variational techniques, WKB approximation	To be able to do simple calculations with time independent perturbation theory, simple calculations with variational and WKB methods.	TB: Ch. 5, RB1: Ch 16, 17.
29-40	Time dependent quantum systems and introduction to quantum theory of scattering	The Interaction picture, Dyson series, application of perturbative techniques to 2 level systems, Fermi Golden rule, Introduction to quantum scattering, Born series, partial waves.	To be able to do simple calculations with time dependent systems.	TB: Ch 5, 6 RB1: Ch 18, 19

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Mins.	30% (90 Marks)	TBA	Open/Closed
Comprehensive Examination	180 Mins.	40% (120 Marks)	TBA	Open/Closed
Tutorial Tests (25 marks each; best 3 out of 4) and one homework assignment (15 marks)	In class/ take home	30% (90 Marks)	TBA	Open/Closed

7. Chamber Consultation Hour: To be announced in the class.

8. Notices: Notices will be posted on Google Classroom/ over email communications.

9. Make-up Policy: **Strictly no make-up for the tutorials (buffer will be available).** If the student is unable to appear for the Regular Test/Examination due to **genuine exigencies****, the student must refer to the procedure for applying for the Make-up Test/Examination. **[**Sickness leading to hospitalization and emergency need to be out of station with prior permission from/intimation to the IC]**

10. Note (if any): It is the responsibility of each student to regularly attend lectures. The handout provides general information about the course, but if additional relevant topics need to be covered for the course's completeness, they will be discussed in class. All class materials may not be available in a single textbook or reference book. Therefore, students are advised to stay up to date with the topics discussed in class.

Arpan Das
Instructor-in-charge
Course No. PHY F311